

Home Work 5 Solutions
II B. Tech. (Common)
Mathematics-II (NMCI102)

Notations:

$M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$M_n(\mathbb{F})$: The set of all matrices of order $n \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.

$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.

1. Examine whether T is a linear transformation. If T is linear, find $\text{Ker}(T)$ and $\text{Im}(T)$, verify the rank-nullity theorem for T , and determine whether T is one-to-one, onto, or both.

- (a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, -a_2)$
- (b) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, 0)$
- (c) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x + 2y, 2x + y, x + 2y)$
- (d) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x - y, 2z)$
- (e) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (yz, zx, xy)$
- (f) $T: M_{m \times n}(\mathbb{R}) \rightarrow M_{n \times m}(\mathbb{R})$ defined by $T(A) = A^t$
- (g) $T: M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R})$ defined by $T(A) = \frac{1}{2}(A + A^t)$
- (h) $T: P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ defined by $T(f(x)) = f'(x)$
- (i) $T: P_2(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(f(x)) = \int_a^b f(t)dt$, where $a, b \in \mathbb{R}$ and $a < b$
- (j) $T: P_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ defined by $T(a_0 + a_1x + a_2x^2) = (a_0, a_1, a_2)$
- (k) $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ defined by $T(f(x)) = 2f'(x) + \int_0^x 3f(t)dt$
- (l) $T: P_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ defined by $T(f(x)) = \begin{bmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{bmatrix}$
- (m) $T: M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \text{trace}(A)$
- (n) $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ defined by $T(f(x)) = \int_0^x f(t)dt$
- (o) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (1, a_2)$
- (p) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, a_1^2)$
- (q) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (\sin a_1, 0)$
- (r) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (|a_1|, a_2)$
- (s) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1 + 1, a_2)$.

Solutions:

(a). 1. Linearity

Let $u = (a_1, a_2)$, $v = (b_1, b_2)$, and $c \in \mathbb{R}$.

Additivity:

$$T(u + v) = T(a_1 + b_1, a_2 + b_2) = (a_1 + b_1, -(a_2 + b_2)) = (a_1, -a_2) + (b_1, -b_2) = T(u) + T(v).$$

Homogeneity:

$$T(cu) = T(ca_1, ca_2) = (ca_1, -ca_2) = c(a_1, -a_2) = cT(u).$$

Since both properties hold, T is linear.

2. Kernel

To find the kernel, solve

$$T(a_1, a_2) = (0, 0).$$

$$(a_1, -a_2) = (0, 0).$$

Thus,

$$a_1 = 0, \quad -a_2 = 0 \Rightarrow a_2 = 0.$$

Hence,

$$\ker(T) = \{(0, 0)\}.$$

Therefore,

$$\text{nullity}(T) = \dim(\ker T) = 0.$$

3. Image

Since $\{(1, 0), (0, 1)\}$ is a basis for $\mathbb{R}^2$ and T is linear,

$$\text{Im}(T) = \text{span}\{T(1, 0), T(0, 1)\}.$$

Compute:

$$T(1, 0) = (1, 0), \quad T(0, 1) = (0, -1).$$

Thus,

$$\text{Im}(T) = \text{span}\{(1, 0), (0, -1)\}.$$

Check linear independence:

Suppose

$$a(1, 0) + b(0, -1) = (0, 0).$$

Then

$$(a, -b) = (0, 0),$$

which implies $a = 0$ and $b = 0$.

Thus the vectors are linearly independent. Since there are two linearly independent vectors in $\mathbb{R}^2$, they form a basis. Therefore,

$$\text{Im}(T) = \mathbb{R}^2.$$

Hence,

$$\text{rank}(T) = 2.$$

4. Rank-Nullity Theorem

$$\dim(\mathbb{R}^2) = \text{rank}(T) + \text{nullity}(T).$$

$$2 = 2 + 0.$$

The theorem is verified.

5. One-to-One and Onto

Since $\ker(T) = \{(0, 0)\}$, T is one-to-one.

Since $\text{Im}(T) = \mathbb{R}^2$, T is onto.

Conclusion.

T is a linear transformation.

$$\ker(T) = \{(0, 0)\}, \quad \text{Im}(T) = \mathbb{R}^2.$$

The Rank–Nullity Theorem holds, and T is both one-to-one and onto (bijective).

(b) 1. Linearity

Let $u = (a_1, a_2)$, $v = (b_1, b_2)$, and $c \in \mathbb{R}$.

Additivity:

$$T(u + v) = T(a_1 + b_1, a_2 + b_2) = (a_1 + b_1, 0) = (a_1, 0) + (b_1, 0) = T(u) + T(v).$$

Homogeneity:

$$T(cu) = T(ca_1, ca_2) = (ca_1, 0) = c(a_1, 0) = cT(u).$$

Since both properties hold, T is linear.

2. Kernel

To find the kernel, solve

$$T(a_1, a_2) = (0, 0).$$

$$(a_1, 0) = (0, 0).$$

Thus,

$$a_1 = 0,$$

while a_2 is arbitrary.

Hence,

$$\ker(T) = \{(0, a_2) \mid a_2 \in \mathbb{R}\} = \text{span}\{(0, 1)\}.$$

Therefore,

$$\text{nullity}(T) = 1.$$

3. Image

Since $\{(1, 0), (0, 1)\}$ is a basis for $\mathbb{R}^2$ and T is linear,

$$\text{Im}(T) = \text{span}\{T(1, 0), T(0, 1)\}.$$

Compute:

$$T(1, 0) = (1, 0), \quad T(0, 1) = (0, 0).$$

Thus,

$$\text{Im}(T) = \text{span}\{(1, 0)\}.$$

Hence,

$$\text{Im}(T) = \{(x, 0) \mid x \in \mathbb{R}\}.$$

Therefore,

$$\text{rank}(T) = 1.$$

4. Rank–Nullity Theorem

$$\dim(\mathbb{R}^2) = \text{rank}(T) + \text{nullity}(T).$$

$$2 = 1 + 1.$$

The theorem is verified.

5. One-to-One and Onto

Since $\ker(T) \neq \{(0, 0)\}$, T is **not one-to-one**.

Since $\text{Im}(T) \neq \mathbb{R}^2$, T is **not onto**.

(c) 1. Linearity

Each component of $T(x, y)$ is a linear combination of x and y , so T is linear.

2. Kernel

Solve $T(x, y) = (0, 0, 0)$:

$$x + 2y = 0, \quad 2x + y = 0.$$

This implies

$$x = y = 0.$$

Thus,

$$\ker(T) = \{(0, 0)\}.$$

So,

$$\text{nullity}(T) = 0.$$

3. Image

Use the standard basis of $\mathbb{R}^2$:

$$T(1, 0) = (1, 2, 1), \quad T(0, 1) = (2, 1, 2).$$

Thus,

$$\text{Im}(T) = \text{span}\{(1, 2, 1), (2, 1, 2)\}.$$

Check linear independence:

Suppose

$$a(1, 2, 1) + b(2, 1, 2) = (0, 0, 0).$$

This gives

$$a + 2b = 0, \quad 2a + b = 0.$$

Solving yields $a = b = 0$, so the vectors are linearly independent.

Hence,

$$\text{rank}(T) = 2.$$

Since the image has dimension 2 in $\mathbb{R}^3$, it is a 2-dimensional subspace of $\mathbb{R}^3$.

4. Rank–Nullity

$$\dim(\mathbb{R}^2) = \text{rank}(T) + \text{nullity}(T)$$

$$2 = 2 + 0.$$

Verified.

5. One-to-One and Onto

Since $\ker(T) = \{(0, 0)\}$, T is one-to-one.

Since $\text{Im}(T)$ is 2-dimensional in $\mathbb{R}^3$ (dimension 3), T is not onto.

(d) 1. Linearity

Each component is a linear combination of x, y, z , so T is linear.

2. Kernel

Solve

$$T(x, y, z) = (0, 0).$$

$$x - y = 0, \quad 2z = 0.$$

Thus,

$$x = y, \quad z = 0.$$

Let $x = t$. Then $y = t$.

$$\ker(T) = \{(t, t, 0) \mid t \in \mathbb{R}\} = \text{span}\{(1, 1, 0)\}.$$

Hence,

$$\text{nullity}(T) = 1.$$

3. Image

Use the standard basis of $\mathbb{R}^3$:

$$T(1, 0, 0) = (1, 0), \quad T(0, 1, 0) = (-1, 0), \quad T(0, 0, 1) = (0, 2).$$

Thus,

$$\text{Im}(T) = \text{span}\{(1, 0), (0, 2)\}.$$

These two vectors are linearly independent, so

$$\text{Im}(T) = \mathbb{R}^2.$$

Hence,

$$\text{rank}(T) = 2.$$

4. Rank–Nullity

$$\begin{aligned} \dim(\mathbb{R}^3) &= \text{rank}(T) + \text{nullity}(T) \\ 3 &= 2 + 1. \end{aligned}$$

Verified.

5. One-to-One and Onto

Since $\ker(T) \neq \{0\}$, T is not one-to-one.

Since $\text{Im}(T) = \mathbb{R}^2$, T is onto.

(e) To determine whether T is linear, we check whether it satisfies additivity and homogeneity.

Homogeneity Test

Let $c \in \mathbb{R}$. Then

$$T(cx, cy, cz) = ((cy)(cz), (cz)(cx), (cx)(cy)) = (c^2yz, c^2zx, c^2xy).$$

On the other hand,

$$cT(x, y, z) = c(yz, zx, xy) = (cyz, czx, cxy).$$

Since

$$(c^2yz, c^2zx, c^2xy) \neq (cyz, czx, cxy)$$

for general c , the homogeneity property fails. Since T does not satisfy the homogeneity property, it is **not a linear transformation**.

(f) 1. Linearity

Let $A, B \in M_{m \times n}(\mathbb{R})$ and $c \in \mathbb{R}$.

$$T(A + B) = (A + B)^t = A^t + B^t = T(A) + T(B),$$

$$T(cA) = (cA)^t = cA^t = cT(A).$$

Hence T is linear.

2. Kernel

$$T(A) = 0 \iff A^t = 0.$$

This happens only when $A = 0$. Thus,

$$\ker(T) = \{0\}.$$

Therefore, $\text{nullity}(T) = 0$.

3. Image

For any $B \in M_{n \times m}(\mathbb{R})$, let $A = B^t$. Then

$$T(A) = A^t = (B^t)^t = B.$$

Thus every matrix in $M_{n \times m}(\mathbb{R})$ is in the image, so

$$\text{Im}(T) = M_{n \times m}(\mathbb{R}).$$

Hence,

$$\text{rank}(T) = mn.$$

4. Rank-Nullity

$$\dim M_{m \times n}(\mathbb{R}) = mn.$$

$$mn = \text{rank}(T) + \text{nullity}(T) = mn + 0.$$

Verified.

$T(A) = A^t$ is linear, one-to-one, and onto.

(g) 1. Linearity

Let $A, B \in M_{n \times n}(\mathbb{R})$ and $c \in \mathbb{R}$.

$$T(A + B) = \frac{1}{2}((A + B) + (A + B)^t) = \frac{1}{2}(A + A^t + B + B^t) = T(A) + T(B),$$

$$T(cA) = \frac{1}{2}(cA + (cA)^t) = \frac{1}{2}(cA + cA^t) = cT(A).$$

Thus T is linear.

2. Kernel

$$T(A) = 0 \iff \frac{1}{2}(A + A^t) = 0 \iff A + A^t = 0.$$

Thus

$$A^t = -A.$$

Hence,

$$\ker(T) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^t = -A\},$$

the set of skew-symmetric matrices.

The dimension of this space is

$$\frac{n(n-1)}{2}.$$

3. Image

For any A ,

$$T(A) = \frac{1}{2}(A + A^t).$$

Notice

$$T(A)^t = \frac{1}{2}(A^t + A) = T(A),$$

so every image matrix is symmetric.

Conversely, if S is symmetric ($S^t = S$), then

$$T(S) = \frac{1}{2}(S + S^t) = \frac{1}{2}(2S) = S.$$

Thus,

$$\text{Im}(T) = \{S \in M_{n \times n}(\mathbb{R}) \mid S^t = S\},$$

the set of symmetric matrices.

The dimension of this space is

$$\frac{n(n+1)}{2}.$$

4. Rank-Nullity

$$\dim M_{n \times n}(\mathbb{R}) = n^2.$$

$$n^2 = \frac{n(n+1)}{2} + \frac{n(n-1)}{2}.$$

Verified.

Conclusion: $T(A) = \frac{1}{2}(A + A^t)$ is linear.

$$\ker(T) = \{\text{skew-symmetric matrices}\},$$

$$\text{Im}(T) = \{\text{symmetric matrices}\}.$$

It is neither one-to-one nor onto (unless $n = 0$).

(h) 1. Linearity

Let $f(x), g(x) \in P_n(\mathbb{R})$ and $c \in \mathbb{R}$.

$$T(f + g) = (f + g)' = f' + g' = T(f) + T(g),$$

$$T(cf) = (cf)' = cf' = cT(f).$$

Hence differentiation satisfies additivity and homogeneity, so T is linear.

2. Kernel

The kernel consists of all polynomials whose derivative is zero:

$$T(f) = 0 \iff f'(x) = 0.$$

This happens exactly when $f(x)$ is a constant polynomial.

Thus,

$$\ker(T) = \{c \mid c \in \mathbb{R}\} = \text{span}\{1\}.$$

Therefore,

$$\text{nullity}(T) = 1.$$

3. Image

Let $g(x) \in P_{n-1}(\mathbb{R})$. Then there exists a polynomial

$$f(x) = \int g(x) dx$$

of degree at most n such that $f'(x) = g(x)$.

Hence every polynomial in $P_{n-1}(\mathbb{R})$ is the derivative of some polynomial in $P_n(\mathbb{R})$. Therefore,

$$\text{Im}(T) = P_{n-1}(\mathbb{R}).$$

So,

$$\text{rank}(T) = \dim P_{n-1}(\mathbb{R}) = n.$$

4. Rank–Nullity Theorem

$$\dim P_n(\mathbb{R}) = n + 1.$$

$$n + 1 = \text{rank}(T) + \text{nullity}(T) = n + 1.$$

Verified.

5. One-to-One and Onto

Since $\ker(T) \neq \{0\}$, T is **not one-to-one**.

Since $\text{Im}(T) = P_{n-1}(\mathbb{R})$, T is **onto**.

(i) 1. Linearity

Let $f(x), g(x) \in P_2(\mathbb{R})$ and $c \in \mathbb{R}$. Then

$$T(f+g) = \int_a^b (f(t) + g(t)) dt = \int_a^b f(t) dt + \int_a^b g(t) dt = T(f) + T(g),$$

$$T(cf) = \int_a^b cf(t) dt = c \int_a^b f(t) dt = cT(f).$$

Thus T is linear.

2. Kernel

The kernel consists of all polynomials $f \in P_2(\mathbb{R})$ such that

$$T(f) = \int_a^b f(t) dt = 0.$$

Let $f(x) = \alpha + \beta x + \gamma x^2$. Then

$$\int_a^b f(t) dt = \int_a^b (\alpha + \beta t + \gamma t^2) dt = \alpha(b-a) + \frac{\beta}{2}(b^2 - a^2) + \frac{\gamma}{3}(b^3 - a^3).$$

Thus the kernel is the set of all $(\alpha, \beta, \gamma) \in \mathbb{R}^3$ satisfying

$$\alpha(b-a) + \frac{\beta}{2}(b^2 - a^2) + \frac{\gamma}{3}(b^3 - a^3) = 0.$$

This is a plane in $\mathbb{R}^3$, so

$$\dim(\ker(T)) = 2.$$

3. Image

Since $T(f) \in \mathbb{R}$ and $T(1) = \int_a^b 1 dt = b-a \neq 0$, the image contains all real multiples of $b-a$. Therefore,

$$\text{Im}(T) = \mathbb{R}.$$

Hence,

$$\text{rank}(T) = 1.$$

4. Rank-Nullity Theorem

$$\dim P_2(\mathbb{R}) = 3 = \text{rank}(T) + \text{nullity}(T) = 1 + 2.$$

Verified.

5. One-to-One and Onto

Since $\dim(\ker(T)) = 2 > 0$, T is **not one-to-one**. Since $\text{Im}(T) = \mathbb{R}$, T is **onto**.

(j) 1. Linearity

Let $f(x) = a_0 + a_1x + a_2x^2$ and $g(x) = b_0 + b_1x + b_2x^2$ in $P_2(\mathbb{R})$, and $c \in \mathbb{R}$. Then

$$T(f + g) = T((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2) = (a_0 + b_0, a_1 + b_1, a_2 + b_2) = T(f) + T(g),$$

$$T(cf) = T(ca_0 + ca_1x + ca_2x^2) = (ca_0, ca_1, ca_2) = cT(f).$$

Hence T is linear.

2. Kernel

The kernel consists of all polynomials mapped to the zero vector:

$$T(a_0 + a_1x + a_2x^2) = (a_0, a_1, a_2) = (0, 0, 0).$$

This gives $a_0 = a_1 = a_2 = 0$, so

$$\ker(T) = \{0\}.$$

Therefore,

$$\text{nullity}(T) = 0.$$

3. Image

For any $(b_0, b_1, b_2) \in \mathbb{R}^3$, choose the polynomial $f(x) = b_0 + b_1x + b_2x^2$. Then

$$T(f) = (b_0, b_1, b_2),$$

so every vector in $\mathbb{R}^3$ is in the image. Hence,

$$\text{Im}(T) = \mathbb{R}^3,$$

and

$$\text{rank}(T) = 3.$$

4. Rank–Nullity Theorem

$$\dim P_2(\mathbb{R}) = 3 = \text{rank}(T) + \text{nullity}(T) = 3 + 0.$$

Verified.

5. One-to-One and Onto

Since $\ker(T) = \{0\}$, T is **one-to-one**. Since $\text{Im}(T) = \mathbb{R}^3$, T is **onto**.

(k) 1. Linearity

Let $f(x), g(x) \in P_2(\mathbb{R})$ and $c \in \mathbb{R}$. Then

$$T(f + g) = 2(f + g)' + \int_0^x 3(f + g)(t) dt = 2f' + 2g' + \int_0^x 3f(t) dt + \int_0^x 3g(t) dt = T(f) + T(g),$$

$$T(cf) = 2(cf)' + \int_0^x 3(cf)(t) dt = 2cf' + \int_0^x 3cf(t) dt = cT(f).$$

Hence T is linear.

2. Kernel

Let $f(x) = a_0 + a_1x + a_2x^2 \in P_2(\mathbb{R})$. Then

$$f'(x) = a_1 + 2a_2x, \quad \int_0^x 3f(t) dt = \int_0^x 3(a_0 + a_1t + a_2t^2) dt = 3a_0x + \frac{3a_1}{2}x^2 + a_2x^3.$$

Therefore,

$$T(f(x)) = 2(a_1 + 2a_2x) + 3a_0x + \frac{3a_1}{2}x^2 + a_2x^3.$$

Simplify component-wise:

$$T(f(x)) = 2a_1 + (4a_2 + 3a_0)x + \frac{3a_1}{2}x^2 + a_2x^3.$$

Set $T(f(x)) = 0$. Then the coefficients of each power of x must vanish. This gives,

$$2a_1 = 0, \quad 4a_2 + 3a_0 = 0, \quad \frac{3a_1}{2} = 0, \quad a_2 = 0.$$

From $a_2 = 0$, the second equation becomes $3a_0 = 0 \implies a_0 = 0$.

Hence

$$\ker(T) = \{0\}.$$

Therefore,

$$\text{nullity}(T) = 0.$$

3. Image

$$T(f(x)) = 2a_1 + (4a_2 + 3a_0)x + \frac{3a_1}{2}x^2 + a_2x^3.$$

Let us express the image as a span of basis vectors. Let $f_1 = 1, f_2 = x, f_3 = x^2 \in P_2(\mathbb{R})$.

$$T(1) = 2 \cdot 0 + \int_0^x 3 \cdot 1 dt = 3x,$$

$$T(x) = 2 \cdot 1 + \int_0^x 3t dt = 2 + \frac{3}{2}x^2,$$

$$T(x^2) = 2 \cdot 2x + \int_0^x 3t^2 dt = 4x + x^3.$$

So

$$\text{Im}(T) = \text{span}\{3x, 2 + \frac{3}{2}x^2, 4x + x^3\}.$$

Check linear independence:

Suppose

$$\alpha(3x) + \beta(2 + \frac{3}{2}x^2) + \gamma(4x + x^3) = 0 \quad \forall x$$

$$\implies \gamma x^3 + \frac{3}{2}\beta x^2 + (3\alpha + 4\gamma)x + 2\beta = 0.$$

By Comparing coefficients:

$$\gamma = 0, \quad \frac{3}{2}\beta = 0, \quad 3\alpha + 4\gamma = 0, \quad 2\beta = 0.$$

This implies

$$\gamma = 0, \quad \beta = 0, \quad \alpha = 0.$$

Thus the set $\{3x, 2 + \frac{3}{2}x^2, 4x + x^3\}$ is linearly independent.

Hence

$$\dim(\text{Im}(T)) = 3.$$

4. Rank–Nullity

$$\dim P_2(\mathbb{R}) = 3 = \text{rank}(T) + \text{nullity}(T) = 3 + 0.$$

Verified.

5. One-to-One and Onto

Since $\ker(T) = \{0\}$, T is **one-to-one**.

Since $\dim(\text{Im}(T)) = 3 = \dim P_2(\mathbb{R}) < \dim P_3(\mathbb{R}) = 4$, T is **not onto**.

(1) Step 1: Linearity

Let $f, g \in P_2(\mathbb{R})$ and $\alpha \in \mathbb{R}$. Then

$$\begin{aligned} T(f+g) &= \begin{bmatrix} (f+g)(1) - (f+g)(2) & 0 \\ 0 & (f+g)(0) \end{bmatrix} = \begin{bmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{bmatrix} + \begin{bmatrix} g(1) - g(2) & 0 \\ 0 & g(0) \end{bmatrix} \\ &\implies T(f+g) = T(f) + T(g), \end{aligned}$$

and

$$T(\alpha f) = \begin{bmatrix} (\alpha f)(1) - (\alpha f)(2) & 0 \\ 0 & (\alpha f)(0) \end{bmatrix} = \alpha \begin{bmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{bmatrix} = \alpha T(f).$$

Hence, T is linear.

Step 2: Kernel

Let $f(x) = ax^2 + bx + c \in P_2(\mathbb{R})$. Then

$$T(f) = 0 \implies \begin{cases} f(1) - f(2) = 0 \\ f(0) = 0 \end{cases} \implies \begin{cases} (a+b+c) - (4a+2b+c) = -3a-b=0 \\ c=0 \end{cases}.$$

Thus $b = -3a$ and $c = 0$, so

$$f(x) = a(x^2 - 3x), \quad a \in \mathbb{R}.$$

Therefore,

$$\ker(T) = \text{span}\{x^2 - 3x\}, \quad \dim(\ker(T)) = 1.$$

Step 3: Range

Consider the standard basis of $P_2(\mathbb{R})$: $\{1, x, x^2\}$. Compute the images under T :

$$T(1) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad T(x) = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \quad T(x^2) = \begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix}.$$

Check for linear independence: suppose

$$\alpha T(1) + \beta T(x) + \gamma T(x^2) = 0 \implies \alpha \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + \beta \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} + \gamma \begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

From the (1,1) entry: $-\beta - 3\gamma = 0 \implies \beta = -3\gamma$, and from the (2,2) entry: $\alpha = 0$.

Since γ is free, the set $\{T(1), T(x), T(x^2)\}$ is linearly dependent. A maximal linearly independent subset is

$$\mathcal{B}_{\text{Im}} = \{T(1), T(x)\} = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \right\}.$$

Hence the range of T is

$$\text{Im}(T) = \text{span}\{T(1), T(x)\} = \left\{ \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} : \alpha, \beta \in \mathbb{R} \right\}, \quad \dim(\text{Im}(T)) = 2.$$

Step 4: Rank-Nullity Theorem

$$\dim(P_2(\mathbb{R})) = 3 = \dim(\ker(T)) + \dim(\text{Im}(T)) = 1 + 2.$$

Step 5: One-to-One and Onto

$\ker(T) \neq \{0\}$, so T is not injective. $\dim(\text{Im}(T)) = 2 < \dim(M_{2 \times 2}(\mathbb{R})) = 4$, so T is not surjective.

(m) Step 1: Linearity

Let $A, B \in M_{2 \times 2}(\mathbb{R})$ and $\alpha \in \mathbb{R}$. Then

$$T(A + B) = \text{trace}(A + B) = \text{trace}(A) + \text{trace}(B) = T(A) + T(B),$$

and

$$T(\alpha A) = \text{trace}(\alpha A) = \alpha \text{trace}(A) = \alpha T(A).$$

Hence, T is linear.

Step 2: Kernel

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}).$$

Then $T(A) = a + d$. For $A \in \ker(T)$, we require $a + d = 0 \implies d = -a$. Therefore,

$$\ker(T) = \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} : a, b, c \in \mathbb{R} \right\}.$$

Dimension: $\dim(\ker(T)) = 3$ (parameters a, b, c are free).

Step 3: Image

For any $r \in \mathbb{R}$, choose

$$A = \begin{bmatrix} r & 0 \\ 0 & 0 \end{bmatrix} \implies T(A) = r.$$

Hence,

$$\text{Im}(T) = \mathbb{R}, \quad \dim(\text{Im}(T)) = 1.$$

Step 4: Rank-Nullity

$$\dim(M_{2 \times 2}(\mathbb{R})) = 4, \quad \dim(\ker(T)) = 3, \quad \dim(\text{Im}(T)) = 1.$$

Rank-nullity theorem: $\dim(M_{2 \times 2}(\mathbb{R})) = 4 = 3 + 1 = \dim(\ker(T)) + \dim(\text{Im}(T))$.

Step 5: One-to-One and Onto

Since $\ker(T) \neq \{0\}$, T is not One-to-one. Since $\text{Im}(T) = \mathbb{R}$, T is onto.

(n) Step 1: Linearity

Let $f, g \in P_2(\mathbb{R})$ and $\alpha \in \mathbb{R}$. Then

$$T(f + g)(x) = \int_0^x (f(t) + g(t)) dt = \int_0^x f(t) dt + \int_0^x g(t) dt = T(f)(x) + T(g)(x),$$

and

$$T(\alpha f)(x) = \int_0^x \alpha f(t) dt = \alpha \int_0^x f(t) dt = \alpha T(f)(x).$$

Hence, T is linear.

Step 2: Kernel

Let $f(x) = ax^2 + bx + c \in P_2(\mathbb{R})$. Then

$$T(f)(x) = \int_0^x (at^2 + bt + c) dt = \frac{a}{3}x^3 + \frac{b}{2}x^2 + cx.$$

We require $T(f)(x) = 0$ for all x . Therefore,

$$\frac{a}{3}x^3 + \frac{b}{2}x^2 + cx = 0 \quad \forall x \implies a = 0, b = 0, c = 0.$$

Thus,

$$\ker(T) = \{0\}.$$

Step 3: Image

For $f(x) = ax^2 + bx + c$, we have

$$T(f)(x) = \frac{a}{3}x^3 + \frac{b}{2}x^2 + cx.$$

Notice that every polynomial in the image has no constant term, so

$$\text{Im}(T) = \{p(x) \in P_3(\mathbb{R}) : p(0) = 0\} = \text{span}\{x, x^2, x^3\}.$$

Hence,

$$\dim(\text{Im}(T)) = 3.$$

Step 4: Rank-Nullity

$$\dim(P_3(\mathbb{R})) = 3 = \dim(\ker(T)) + \dim(\text{Im}(T)) = 0 + 3.$$

Step 5: One-to-One and Onto

Since $\ker(T) = \{0\}$, T is One-to-One. Since $\dim(\text{Im}(T)) = 3 < \dim(P_3(\mathbb{R})) = 4$, T is not onto.

(o) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(a_1, a_2) = (1, a_2)$

Not linear (constant term 1).

(p) $T(a_1, a_2) = (a_1, a_1^2)$

Not linear (square term).

(q) $T(a_1, a_2) = (\sin a_1, 0)$

Not linear (sine).

(r) $T(a_1, a_2) = (|a_1|, a_2)$

Not linear (absolute value).

(s) $T(a_1, a_2) = (a_1 + 1, a_2)$

Not linear (constant shift).

2. (a) Is there a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(1, 0) = (1, 4)$ and $T(1, 1) = (2, 5)$? If yes, what is $T(2, 3)$?

Solutions: Yes, such a linear transformation exists.

Since $(1, 0)$ and $(1, 1)$ form a basis of $\mathbb{R}^2$, the transformation is uniquely determined.

First find $T(0, 1)$:

$$(0, 1) = (1, 1) - (1, 0).$$

$$T(0, 1) = T(1, 1) - T(1, 0) = (2, 5) - (1, 4) = (1, 1).$$

Now for any (x, y) :

$$T(x, y) = xT(1, 0) + yT(0, 1) = x(1, 4) + y(1, 1) = (x + y, 4x + y).$$

Finally,

$$T(2, 3) = (2 + 3, 4(2) + 3) = (5, 11).$$

- (b) Is there a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that $T(1, 1) = (1, 0, 2)$ and $T(2, 3) = (1, -1, 4)$?
If yes, find T and what is $T(8, 11)$?

Solutions: Yes, such a linear transformation exists.

Since $(1, 1)$ and $(2, 3)$ are linearly independent, they form a basis of $\mathbb{R}^2$, so T is uniquely determined.

Write (x, y) as

$$(x, y) = a(1, 1) + b(2, 3).$$

Then

$$a + 2b = x, \quad a + 3b = y.$$

Subtracting gives $b = y - x$ and

$$a = 3x - 2y.$$

Thus

$$T(x, y) = (3x - 2y)T(1, 1) + (y - x)T(2, 3).$$

$$= (3x - 2y)(1, 0, 2) + (y - x)(1, -1, 4).$$

$$T(x, y) = (2x - y, x - y, 2x).$$

Finally,

$$T(8, 11) = (2(8) - 11, 8 - 11, 2(8)) = (5, -3, 16).$$

- (c) Is there a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(1, 0, 3) = (1, 1)$ and $T(-2, 0, -6) = (2, 1)$?

Solutions: No, such a linear transformation does not exist.

Notice that

$$(-2, 0, -6) = -2(1, 0, 3).$$

If T were linear, then

$$T(-2, 0, -6) = T(-2(1, 0, 3)) = -2T(1, 0, 3).$$

$$= -2(1, 1) = (-2, -2).$$

But we are given that

$$T(-2, 0, -6) = (2, 1).$$

Since $(-2, -2) \neq (2, 1)$, this contradicts linearity.

Therefore, no such linear transformation exists.

- (d) Determine the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ which maps the basis vectors $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$ of $\mathbb{R}^3$ to $(1, 1, 1)$, $(1, 1, 1)$, $(1, 1, 1)$ respectively. Verify rank-nullity theorem after finding $\text{Ker}(T)$, $\text{Im}(T)$.

Solutions: Since

$$v_1 = (0, 1, 1), \quad v_2 = (1, 0, 1), \quad v_3 = (1, 1, 0)$$

form a basis of $\mathbb{R}^3$ (their determinant is $2 \neq 0$), there exists a unique linear transformation T .

Write

$$(x, y, z) = av_1 + bv_2 + cv_3.$$

Then

$$(x, y, z) = (b + c, a + c, a + b).$$

Solving,

$$a = \frac{y + z - x}{2}, \quad b = \frac{x + z - y}{2}, \quad c = \frac{x + y - z}{2}.$$

Since

$$T(v_1) = T(v_2) = T(v_3) = (1, 1, 1),$$

we get

$$T(x, y, z) = (a + b + c)(1, 1, 1).$$

$$a + b + c = \frac{x + y + z}{2}.$$

Therefore,

$$T(x, y, z) = \frac{x + y + z}{2}(1, 1, 1).$$

Kernel:

$$\ker(T) = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}.$$

$$\dim(\ker T) = 2.$$

Image:

$$\operatorname{Im}(T) = \operatorname{span}\{(1, 1, 1)\}.$$

$$\dim(\operatorname{Im} T) = 1.$$

Rank–Nullity:

$$2 + 1 = 3 = \dim(\mathbb{R}^3).$$

3. Find all linear transformations $T: \mathbb{F} \rightarrow \mathbb{F}$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Solutions: Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Since $\mathbb{F}$ is a 1–dimensional vector space over itself, a basis is $\{1\}$.

Let

$$T(1) = a \in \mathbb{F}.$$

For any $x \in \mathbb{F}$,

$$x = x \cdot 1.$$

By linearity,

$$T(x) = T(x \cdot 1) = xT(1) = ax.$$

Therefore every linear transformation

$$T: \mathbb{F} \rightarrow \mathbb{F}$$

is of the form

$$T(x) = ax \quad \text{for some } a \in \mathbb{F}.$$

If $a = 0$:

$$\ker(T) = \mathbb{F}, \quad \operatorname{Im}(T) = \{0\}.$$

If $a \neq 0$:

$$\ker(T) = \{0\}, \quad \operatorname{Im}(T) = \mathbb{F}.$$

4. Find all linear transformations $T: \mathbb{F}^2 \rightarrow \mathbb{F}^2$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Solutions: Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

The standard basis of $\mathbb{F}^2$ is

$$e_1 = (1, 0), \quad e_2 = (0, 1).$$

Let

$$T(1, 0) = (a, c), \quad T(0, 1) = (b, d),$$

where $a, b, c, d \in \mathbb{F}$.

For any $(x, y) \in \mathbb{F}^2$,

$$(x, y) = x(1, 0) + y(0, 1).$$

By linearity,

$$T(x, y) = xT(1, 0) + yT(0, 1) = x(a, c) + y(b, d) = (ax + by, cx + dy).$$

Thus every linear transformation

$$T: \mathbb{F}^2 \rightarrow \mathbb{F}^2$$

is of the form

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad a, b, c, d \in \mathbb{F}.$$

5. A linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(x_1, x_2, x_3) = (3x_1 - 2x_2 + x_3, x_1 - 3x_2 - 2x_3)$. Find the matrix of T relative to the ordered bases
- (a) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$.
- (b) $\{(0, 1, 0), (1, 0, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ and $\{(0, 1), (1, 0)\}$ of $\mathbb{R}^2$.
- (c) $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$.

Solutions: Given

$$T(x_1, x_2, x_3) = (3x_1 - 2x_2 + x_3, x_1 - 3x_2 - 2x_3).$$

(a) Standard bases:

$$[T]_a = \begin{pmatrix} 3 & -2 & 1 \\ 1 & -3 & -2 \end{pmatrix}.$$

(b) Domain basis $\{(0, 1, 0), (1, 0, 0), (0, 0, 1)\}$, codomain basis $\{(0, 1), (1, 0)\}$:

$$[T]_b = \begin{pmatrix} -3 & 1 & -2 \\ -2 & 3 & 1 \end{pmatrix}.$$

(c) Domain basis $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$, standard codomain basis:

$$[T]_c = \begin{pmatrix} -1 & 4 & 1 \\ -5 & -1 & -2 \end{pmatrix}.$$

6. The matrix of a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with respect to the ordered basis $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$ is given by $\begin{bmatrix} 0 & 3 & 0 \\ 2 & 3 & -2 \\ 2 & -1 & 2 \end{bmatrix}$. Find T and also find the matrix of T with respect to the ordered basis $\{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$ of $\mathbb{R}^3$.

Solution:

Step 1: Compute $T(v_i)$ in standard coordinates

Let $v_1 = (0, 1, 1)$, $v_2 = (1, 0, 1)$ and $v_3 = (1, 1, 0)$. By definition, the columns of $[T]_{\mathcal{B}}$ represent $T(v_i)$ in the $\mathcal{B}$ -basis. Therefore:

$$\begin{aligned} T(v_1) &= 0v_1 + 2v_2 + 2v_3 = 2v_2 + 2v_3 = (4, 2, 2), \\ T(v_2) &= 3v_1 + 3v_2 - 1v_3 = 3v_1 + 3v_2 - v_3 = (2, 2, 6), \\ T(v_3) &= -2v_2 + 2v_3 = (0, 2, -2). \end{aligned}$$

Step 2: Express an arbitrary vector in $\mathcal{B}$ -coordinates

Let $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ and write

$$x = c_1v_1 + c_2v_2 + c_3v_3.$$

Solving the system, we obtain:

$$c_1 = \frac{x_2 - x_1 + x_3}{2}, \quad c_2 = \frac{x_1 - x_2 + x_3}{2}, \quad c_3 = \frac{x_1 + x_2 - x_3}{2}.$$

Step 3: Compute $T(x)$ in standard coordinates

$$\begin{aligned} T(x) &= c_1T(v_1) + c_2T(v_2) + c_3T(v_3) \\ &= (-x_1 + x_2 + 3x_3, x_1 + x_2 + x_3, x_1 - 3x_2 + 5x_3). \end{aligned}$$

Step 4: Compute $T(w_i)$ in standard coordinates

$$\begin{aligned} T(w_1) &= T(2, 1, 1) = (2, 4, 4), \\ T(w_2) &= T(1, 2, 1) = (4, 4, 0), \\ T(w_3) &= T(1, 1, 2) = (6, 4, 8). \end{aligned}$$

Step 5: Express $T(w_i)$ in $\mathcal{C}$ -coordinates

Solve $T(w_i) = a_i w_1 + b_i w_2 + c_i w_3$ for each i :

$$T(w_1) = -\frac{1}{2}w_1 + \frac{3}{2}w_2 + \frac{3}{2}w_3,$$

$$T(w_2) = 2w_1 + 2w_2 - 2w_3,$$

$$T(w_3) = \frac{3}{2}w_1 - \frac{1}{2}w_2 + \frac{7}{2}w_3.$$

Step 6: Matrix of T with respect to $\mathcal{C}$

The columns of $[T]_{\mathcal{C}}$ are the coordinates of $T(w_i)$ in the $\mathcal{C}$ -basis:

$$[T]_{\mathcal{C}} = \begin{bmatrix} -1/2 & 2 & 3/2 \\ 3/2 & 2 & -1/2 \\ 3/2 & -2 & 7/2 \end{bmatrix}.$$